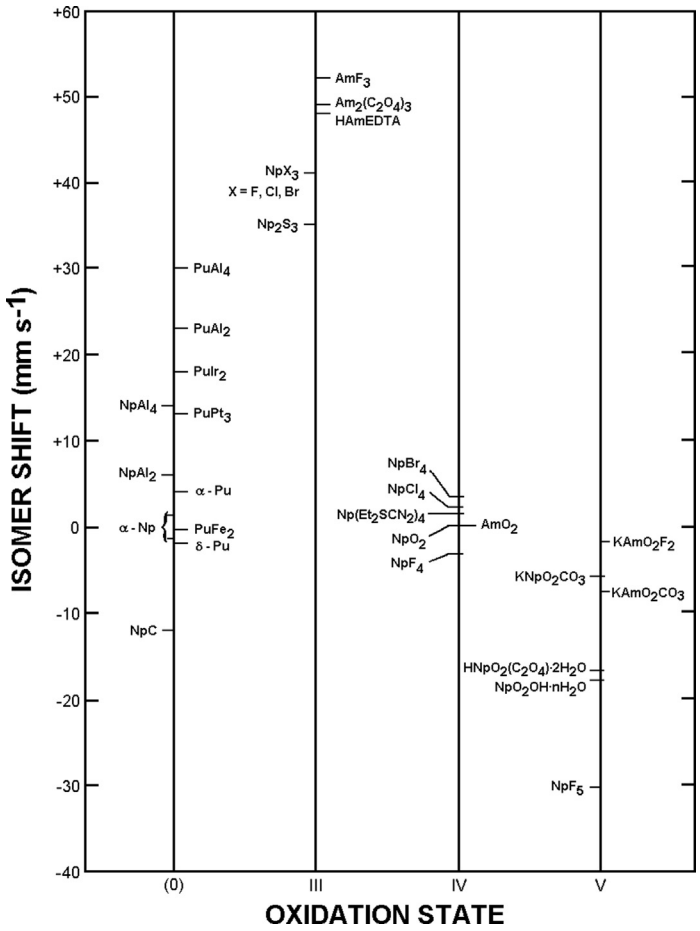


Figure 7.27 Isomer shifts for some actinide metals and actinide compounds.

7.9. TECHNICAL APPLICATIONS OF RADIATION SOURCES

Nuclear radiation absorption methods have many technical applications. These methods are not to be confused with radioisotope tracer methods, although radioisotopes may be used as radiation sources. In the tracer method the chemical properties of the radionuclide are important while in the applications discussed in this section only the type and energy of radiation emitted are important.

As a *source* of radiation in such technical applications, either accelerators or radiation from radionuclides can be used. Interchangeable radionuclides have the advantage over accelerators as radiation sources in that they can cover a larger energy range from high energy γ-rays to low energy β-rays in a much simpler way. This makes it possible to select the type and energy of radiation which have the most advantageous properties for a

particular use. An additional advantage of radionuclides is that the sources can usually be made much smaller than X-ray sources, enabling them to be used in places where larger equipment is inconvenient or impossible to place. The fact that radionuclides require neither electric power nor cooling also renders them more suitable for field applications. Further, their independence from effects of temperature, pressure, and many other factors results in higher reliability compared to X-ray generators. Finally, and, perhaps as important as any of the other factors, they are in general much less expensive than accelerators.

To counterbalance these advantages is the disadvantage of the inability to turn off the radiation from radionuclides. This often requires that the radiation source be well shielded, adding to its weight and cost. An additional drawback to the use of radionuclides is that they have to be replaced after a few half-lives. The seriousness of this disadvantage depends upon the lifetime of the particular nuclide and is unimportant in cases where long-lived sources can be used.

The extensive use of radionuclides in industry is illustrated by Table 7.3, which summarizes the various studies undertaken by I.C.I., UK, in "a typical year".

7.9.1. Radionuclide gauges

Radionuclide gauges are measurement systems consisting of two parts, a radioactive source and a detector, fixed in some geometry to each other. They are used mainly for control in

Table 7.3 Radioisotope based studies undertaken annually by a large chemical company

Technique[(a)]	Number of applications
Level and interface measurements:	
γ-ray absorption	210
Neutron backscatter	480
γ-ray backscatter (storage cavities)	71
Blockage detection and deposition:	
γ-ray absorption	132
Neutron backscatter	129
Entrainment and voidage:	
γ-ray absorption	86
Thickness and corrosion measurements	15
Distillation column scans	108
Flow measurements:	
Pulse velocity	483
Dilution techniques	84
Leak detection	90
Residence-time studies	21
Carry-over studies (tracer)	6

[(a)]Less commonly used techniques have not been included.

industrial processes but can also be applied for specific analyses. The gauges come in two types. In one type the radiation source and the detector are on opposite sides of the technical arrangement to be measured; these are known as transmission or absorption instruments. In the second type, known as reflection or back-scattering instruments, the radiation source and the detector are on the same side. The instruments are also classified with respect to the kind of radiation involved. For example, γ-transmission, β-reflection, secondary X-ray instruments, etc. These radioisotope gauges are used for measurements of thicknesses, densities, etc. In Figure 7.28 a number of types of application for transmission instruments are illustrated. Illustrations a–c show measurements of level control. Type a can be used only for liquids while b and c can be used for all kinds of material. The latter instruments are uniquely suitable for application to large storage containers for grain, wood chips, oil, sand, cement, etc., and for material under extreme conditions such as molten glass and metal, explosives, etc. Level gauges are also used in the control of the filling or packages, cans, etc. in industry as illustrated in d.

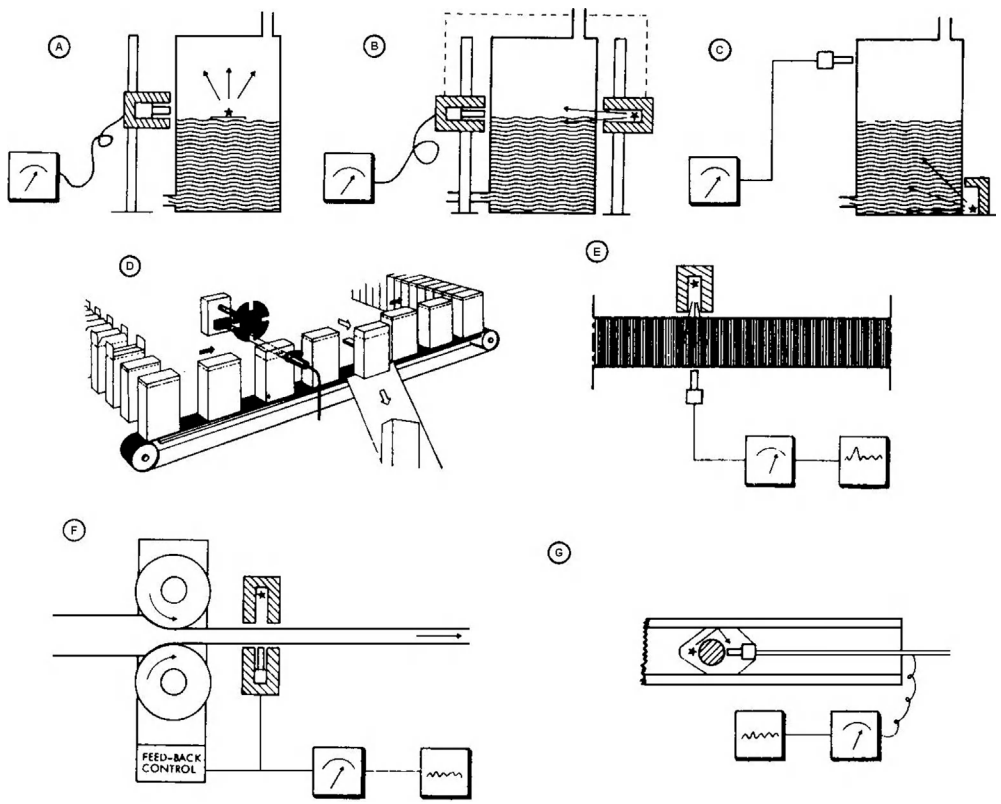


Figure 7.28 Radionuclide gauges for: A-C level measurements in tanks, D control of package filling, E flow density measurements, F thickness control, and G wall material in pipes and bore holes.

Let us consider a somewhat unusual application of this type of instrument. In the manufacture of titanium it is important to keep liquid $TiCl_4$ at a particular high temperature and pressure in a vessel. $TiCl_4$ has a triple point (i.e. the pressure and temperature conditions at which all three phases - solid, liquid, and vapor - of a substance are at equilibrium) in the neighbourhood of the particular technical conditions. By using a γ-density gauge it is possible to detect when the triple point is exceeded because of the disappearance of the vapor-liquid interface. This allows a simple method of control of the process conditions in the vessel.

The use of radioisotope gauges in density measurements is dependent upon Eqn. (7.27) in which r and x are constant while the absorption coefficient is density dependent (i.e. dependent on the average atomic composition of the absorber). A practical arrangement is illustrated in Figure 7.28(e), where the density of a medium in a pipeline is measured. This medium may be a mixture of gas and liquid such as water and water liquids with different composition and different amounts of dissolved substances as, for example, oil, salts or acids in water, process solutions in general or sludges, and emulsions such as fruit juices, latex emulsions, etc. From the variation in density the concentration and composition may be determined. Such density gauges are also used for control in filling of soft drink bottles and cans and submerged in rivers and lakes for measuring the depth of the bottom silt, etc. Density gauges are used in the production and the fabrication of such diverse products as automobile tires and cigarette packages.

Thickness gauges are the most common type of instrument using the absorption technique. In this case x in Eqn. (7.27) is varied. Measurements can be carried out on all kinds of materials with thicknesses of $\leq 100\ \mathrm{g\,cm^{-2}}$ and is independent of the temperature and of whether the material is stationary or in motion. Figure 7.28(f) illustrates the application of a thickness gauge in a rolling mill where material of constant thickness is produced by using the signal from the detector for control purposes. Thickness gauges are used in the fabrication of glass, metal, paper, plastic, rubber, candy bars, etc. They have been used for measuring the thickness of snow in polar regions, icing on airplane wings, and other applications in which it is necessary to use remote operation.

In order to measure very thin layers of material such as coatings of paint, wax, and plastic films on papers or other material, two thickness gauges are used with a differential coupling so that one detector measures the uncovered and the other the covered or treated portion of the material. Thickness gauges also are used in industry to measure the degree of wear in industrial machinery. For surface measurements of thicknesses $\leq 0.8\ \mathrm{g\,cm^{-2}}$ most thickness gauges use radiation sources with β-emitters while for thicknesses of $0.8-5\ \mathrm{g\,cm^{-2}}$ bremsstrahlung radiation sources are most suitable. For even thicker materials γ-emitters are used.

Use of reflection gauges depends on the fact that the intensity of the scattered radiation under conditions of constant geometry depends on the thickness of the scattering material and its electron density (if β- or γ-sources are used). If neutrons are used the

mass number of the scattering material is of prime importance. The electron density of the scattering material varies with the particular element and the chemical composition. Frequently, it is possible to determine the thickness and the nature of a surface layer by means of β-scattering. Reflection gauges have been applied to on-line analysis of tin-covered iron plates, metal coatings on plastics, paint layers, and to measuring the protective coating inside pipelines, Fig. 7.28(g). In some instances γ-radiation sources are preferred over β-emitters in measurements of material with greater wall thicknesses, particularly when transmission measurements are not feasible. Steel thickness from 1 to 20 cm has been measured with 5% accuracy using backscattering from ^{60}Co or ^{137}Cs sources of ≈20 μCi intensity (≈750 kBq).

Scattering and reflection are dependent on the electron density of the absorber, which is approximately proportional to the value of Z/A. Backscattering of β-particles from organic compounds is therefore very dependent on the hydrogen concentration ($Z/A = 1$) but fairly independent of the concentration of C, N, and O ($Z/A = 0.5$). This has led to the development of sensitive instruments for hydrogen analysis for various organic and water-containing materials. In an instrument using 10 mCi ^{90}Sr (370 MBq), the hydrogen concentration in a 10 ml sample can be determined in 20 min with 0.03% accuracy. This is not only superior to other conventional analytical methods, it also has the advantage of being a non-destructive technique.

Neutrons are slowed down most effectively by light elements (§10.5). As a consequence, neutron scattering can be used for the analyses of light elements, particularly hydrogen. In one type of instrument the radiation source consists of ^{252}Cf which produces fast neutrons (from spontaneous fission), while the detector is sensitive only to slow neutrons. This system is used for studies of ground water and analysis of bore holes in wells, Fig. 7.28(g). These analyses are usually combined with density determinations using a γ-source, thereby making it possible to identify strata of water, oil, coal, etc.

Some properties and uses of commercial radionuclide gauges are listed in Table 7.4.

7.9.2. Radiography

Radiography is a photographic technique in which nuclear radiation is used instead of light. Medical examination and nondestructive industrial testing using X-rays generated by high-vacuum tubes are the most important areas. A number of suitable sources of radioactive nuclides for producing radiograms are given in Table 7.4.

Beta-radiography is only suitable for thin objects and not widely applied. On the other hand, γ-radiography is a common nondestructive test technique in which normally ^{137}Cs or ^{60}Co has been used. γ-radiography has advantages in field use and in detection of sensitive objects. The radiation source can be ^{60}Co, which is normally kept in a portable radiation shielding of 30−60 kg of lead and situated at the end of a rod so that it can be pushed out of the shielding for use. The photographic film is located in a cassette surrounded by amplifying screens.

Table 7.4 Some commercially available radionuclide gauges and γ-sources for radiography (France)

Radiation	Source	$t_{½}$	Application
α	U or ^{226}Ra	Long	Thickness control in manufacturing paper, aluminum; 60 g/m^2
Soft β	^{147}Pm(0.2 MeV)	2.6 y	Thickness control; ≤400 g/m^2
β, soft γ	^{204}Tl(0.8 MeV)	3.8 y	Thickness: 1.10 mm steel, 3−50 mm glass; 8−100 kg/m^2
Hard β	^{144}Ce(3 MeV)	0.78 y	Thickness 1 mm steel; ≤10kg/m^2
X	^{109}Cd(88 keV)	1.24 y	Detection of S-content in hydrocarbons
n,γ	RaBe, ^{137}Cs	30 y	Moisture-density meter for civil engineering and agriculture
γ	^{60}Co(1.3 MeV)	5.3 y	4 MBq source for backscatter on ≤20 mm steel, 0.4−40 GBq for remote level indication
Soft γ	^{192}Ir(0.3 MeV)	74 d	400 GBq, 26 kg: ≤40 mm steel radiography
Medium γ	^{137}Cs(0.7 MeV)	30 y	400 GBq, 45 kg: ≤70 mm steel pipeline inspection
Hard γ	^{60}Co(1.3 MeV)	5.3 y	10 TBq, 900 kg: ≤180 mm steel radiography

The exposure At (GBq hours) required for an optical density ($\hat{D} = \log$ (incident light/transmitted light)) ~ 2 at an absorber (object) thickness x (cm) using a typical industrial X-ray film and a ^{60}Co γ-ray source positioned at a distance of 1 m from the film can be estimated by the expressions

$$\log(At) = 1.068 + 0.135\,x \quad \text{for an iron absorber} \tag{7.31}$$

$$\log(At) = 1.068 + 0.040\,x \quad \text{for an concrete absorber} \tag{7.32}$$

Exercise 7.14 is an example of the use of these expressions.

Gamma-radiography has been used for determining the number of reinforced iron bars in concrete construction, cavities in various kinds of castings (as explosives, plastics or metals), cracks or other defects in turbine blades in airplane parts, detonators in unexploded bombs, welded joints in pressure vessels, distillation towers and pipes, corrosion inside pipes and furnaces, and medical field X-rays, to mention only a few applications. Gamma-radiography is used throughout the world for product control leading to improved working safety and economy.

Because γ-absorption occurs through interaction with the electrons, objects of high atomic numbers show the strongest absorption. By using neutrons instead of γ-rays, the opposite effect is achieved, i.e. low Z objects are most effective in removing neutrons from a beam. This is used in *neutron radiography* in which both reactor neutrons and neutrons from ^{252}Cf sources are applied. Because of the higher neutron flux from the reactor than

from ^{252}Cf sources of normal size (i.e. $\leq$1 mg) the exposure time at the reactor is much shorter. On the other hand, the small size of the ^{252}Cf source offers other conveniences.

7.9.3. Radionuclide power generators

The absorption of radiation leads to an increase in the temperature of the absorber. An example of this is the absorption of the kinetic energy of fission products in nuclear reactor fuel elements which is a main source of the heat production in reactors. The absorption of decay energy of radioactive nuclides in appropriate absorbing material can be used in a similar - albeit more modest - way as an energy source.

Figure 7.29 shows the principles of two different radioisotope power generators, the larger (to the left) is of SNAP-7 type and produces ~60 W, the smaller one produces ~10 mW. The radiation source for the larger generator consists of 15 rods (A) clad with hastelloy and containing approximately 7 kg of $SrTiO_3$ which has approximately 225 000 Ci of ^{90}Sr. This heat source is surrounded by 120 pairs of lead telluride thermoelements (B) and a radiation shield of 8 cm of depleted uranium (C). The whole arrangement is surrounded by a steel cover with cooling fins. The weight of this generator is 2.3 t with dimensions of 0.85 m in length and 0.55 m in diameter. It is estimated that the lifetime of such an energy source is at least 5 y, although the half-life of ^{90}Sr (30 y) promises a longer period. Radionuclide generators in unmanned lighthouses, navigation buoys, automatic weather stations, etc., in sizes up to about 100 W, have been in use in a number of countries, e.g. Japan, Sweden, the UK, the USA, etc. Since no moving parts are involved, these generators need a minimum of service. Their reliability makes them valuable in remote areas like the Arctic regions where several such generators have been installed. ^{90}Sr is the preferred radionuclide for terrestrial uses. However, at present the fear of a possible release of radioactivity have largely led to their recovery, or replacement by other power sources. The power density of ^{90}Sr is relatively high, 0.93 W g^{-1}, as compared to some other possible radionuclides as ^{137}Cs (0.26 W g^{-1}) and ^{238}Pu (0.55 W g^{-1}), but it is lower than ^{244}Cm (2.8 W g^{-1}) and ^{245}Cm (121 W g^{-1}). Recently ^{60}Co has come into use; the heating mainly comes from absorption of the γ's in a uranium shielding.

^{238}Pu has been used as an energy source in space. Several satellites with radioisotope generators of 25 W have been placed in space. The Apollo project employed a generator "SNAP-27" containing ^{238}Pu with a total weight of 14 kg and producing 50 W power. The Viking landers on the planet Mars used ^{238}Pu as the main energy source – a number of so called RTG-generators. RTG-generators were also used to power the Pioneer vehicles which gave us the first close pictures of the planet Jupiter and in the Voyager missions to the outer planets and beyond (at present they are all leaving the solar system, but some still keep radio contact with Earth) where the ^{238}Pu power source produced

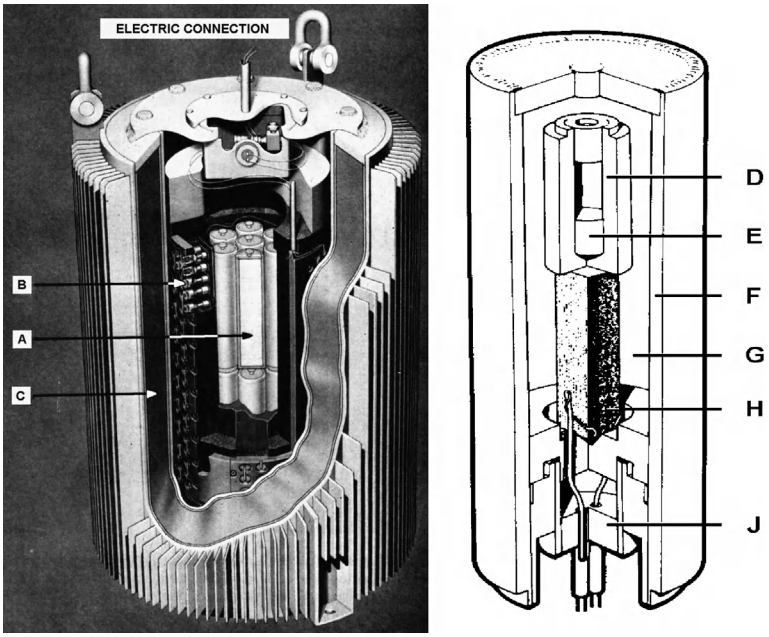


Figure 7.29 A - $^{90}SrTiO_3$; B - thermocouples; C - radiation shield (depleted U); D - capsule; E - ^{238}Pu; F - secondary containment; G - thermal insulation; H - thermopile; J - insulator.

450 W. One will reach the star Sirius in about 300 000 years! In 2012, the most recent RTG use as a power source in space vehicles is on the New Horizons mission to Pluto and Charon.

7.10. EXERCISES

7.1. In §7.3 two equations are given for calculating the range of α-particles in air and in other material from the particle energy as well as a curve for the range in aluminum. How different are the values from the equations and from the curve for a 5 MeV α?

7.2. What is the minimum energy that an α-particle must have to be detected by a GM tube having a mica window of 1.5 mg cm^{-2} (the density of mica is approximately equal to aluminum)?

7.3. For an irradiation experiment it is necessary to extract a beam of deuterons from an accelerator. The projectile energy is 22 MeV D^+. For this purpose the beam is deflected and permitted to pass through a thin titanium foil (density 4.5 g cm^{-3}). Assuming that $\hat{R}_1 \rho_1 M_1^{-½} = \hat{R}_2 \rho_2 M_2^{-½}$ (Bragg-Kleman rule), what is the maximum thickness of the foil? Give the answer in millimeters.